PRISM Benchmark Analysis Report (Rigid / Medium / Difficult)

Consolidated summary of PRISM predictions evaluated against T_Rigid, T_medium, and T_difficult benchmark CSVs using DockQ and iRMSD.

Summary Table

Set	Cx CSV	Models	Match	Unmatch	DQ ok	IR ok	DQ err	IR err
Rigid	162	572	371	201	378	374	21	25
Medium	60	572	134	438	120	118	14	16
Difficult	35	572	67	505	67	67	0	0

Complex-Level Coverage

Set	Cx w/Models	Cx w/DQ	Cx w/IR	Match no DQ	Match no IR
Rigid	110	109	107	21	25
Medium	34	34	33	14	16
Difficult	17	17	17	0	0

Predicted Model Membership by Set

Set	Models Belonging to Set	Percent of All Predicted Models
Rigid	371	64.86%
Medium	134	23.43%
Difficult	67	11.71%

Scope

This report summarizes PRISM prediction analysis against `T_Rigid.csv`, `T_medium.csv`, and `T_difficult.csv` using `DockQ` and `iRMSD`.

Method Logic (Why Scores Are Not Read From PRISM Output Files)

• The target metrics in this analysis are **benchmark comparison metrics** (`DockQ`, `iRMSD`) that require a comparison between:

• a PRISM predicted complex model (`.pdb`) and

• the benchmark native bound complex PDB with a specific chain mapping from the benchmark CSV.

• PRISM output files (e.g. Rosetta score files / intermediate files) do not directly provide these benchmark-comparison metrics with the required benchmark-native chain mapping.

• Therefore, the pipeline recomputes `DockQ` and `iRMSD` from the model PDB + downloaded native bound PDB + CSV-defined chain roles.

• The chain fields reported in `per_prediction.csv` are taken from the model filename (or inferred fallback if omitted), as requested.

Output Locations

• `benchmark/prism_processed_results/prism_rigid_analysis_all_jobs`

• `benchmark/prism_processed_results/prism_medium_analysis_all_jobs`

• `benchmark/prism_processed_results/prism_difficult_analysis_all_jobs`

• Plots: `benchmark/prism_processed_results/benchmark_reports/plots`

Summary Table

```
| Set | Complexes in CSV | Total Models | Matched Models | Unmatched Models | DockQ Scored Rows
| iRMSD Scored Rows | DockQ Errors | iRMSD Errors |
|---|---:|---:|---:|---:|---:|---:|---:|---:|
| Rigid | 162 | 572 | 371 | 201 | 378 | 374 | 21 | 25 |
| Medium | 60 | 572 | 134 | 438 | 120 | 118 | 14 | 16 |
| Difficult | 35 | 572 | 67 | 505 | 67 | 67 | 0 | 0 |
```

Abbreviation guide:

• `DockQ` = DockQ score.

• `iRMSD` = interface RMSD.

• `Scored Rows` = matched model-complex rows where numeric score was produced.

Predicted Model Membership by Set

```
| Set | Models Belonging to Set | Percent of All Predicted Models |
|---|---:|---:|
| Rigid | 371 | 64.86% |
| Medium | 134 | 23.43% |
| Difficult | 67 | 11.71% |
```

Abbreviation guide:

• `Set` = benchmark subset (`Rigid`, `Medium`, `Difficult`).

• `Models Belonging to Set` = unique model files whose PDB ID pair matches at least one row in that set.

Complex-Level Coverage

```
| Set | Complexes With Found Models | Complexes With >=1 DockQ | Complexes With >=1 iRMSD |
Matched Models Missing DockQ | Matched Models Missing iRMSD |
|---|---:|---:|---:|---:|---:|
```

Rigid	110	109	107	21	25
Medium	34	34	33	14	16
Difficult	17	17	17	0	0

Abbreviation guide:

• `Cx` = complexes.

• `DQ` = DockQ.

• `IR` = iRMSD.

• `Match no DQ/IR` = matched rows lacking numeric DockQ/iRMSD output.

Code Logic

1. Match model files to benchmark rows using only `(PDB ID 1, PDB ID 2)` partner names from model filename, ignoring the first template/target token.

2. Assign model partner chains to `PDB ID 1` and `PDB ID 2` roles, including swapped partner-name order handling.

3. Expand each complex row to all native receptor-ligand chain pairs (e.g., `1AHW_AB:C` -> `AC` and `BC`).

4. For each native pair, run DockQ and iRMSD against the downloaded native bound complex PDB.

5. Run reciprocal iRMSD order as well, keep per-model best (`min`) across directions and native pairs.

6. Aggregate by `(PDB ID 1, PDB ID 2, complex)` group: mean/variance/best across predictions/templates.

Key Insights

• Rigid: 371/572 unique models matched the benchmark set (64.9%); DockQ scored 94.7% of matched rows and iRMSD scored 93.7%.

• Medium: 134/572 unique models matched the benchmark set (23.4%); DockQ scored 89.6% of matched rows and iRMSD scored 88.1%.

• Difficult: 67/572 unique models matched the benchmark set (11.7%); DockQ scored 100.0% of matched rows and iRMSD scored 100.0%.

• Best DockQ across all analyzed rows was 0.859 in Rigid (3e1zAB_2nnrA_0_3bpfA_0.rosetta_0001_0001.pdb; pair 2nnr|3bpf, model chains B/A, native pair A:B).

• Best (lowest) iRMSD across all analyzed rows was 0.000 A in Rigid (1xu1BS_1u5yB_0_1xutA_0.rosetta_0001_0001.pdb; pair 1u5y|1xut).

Test Case (1AHW)

• Benchmark row: `Complex=1AHW_AB:C`, `PDB ID 1=1FGN_LH`, `PDB ID 2=1TFH_A`.

• Model A: `1ahwBC_1fgnH_0_1tfhA_0.rosetta_0001_0001.pdb`

• Model chains used: receptor=`H`, ligand=`A`.

• Native pairs evaluated: `AC,BC`.

• DockQ per pair: `AC=0.036409;BC=0.514546`.

• iRMSD forward per pair: `AC=11.022;BC=1.706`.

• iRMSD reverse per pair: `AC=11.022;BC=1.706`.

• Best selected: DockQ=`0.5145464053742148` (Medium), iRMSD=`1.706`.

Core logic used in this case:

```
native_pairs = [('A','C'), ('B','C')]
model_r, model_l = 'H', 'A'
dockq_vals = []
irmsd_vals = []
for ref_r, ref_l in native_pairs:
dockq = DockQ(model, native_1ahw, mapping=f'{model_r}{model_l}:{ref_r}{ref_l}')
ir_fwd = iRMSD(model, model_r, model_l, native_1ahw, ref_r, ref_l)
ir_rev = iRMSD(model, model_l, model_r, native_1ahw, ref_l, ref_r)
dockq_vals.append(dockq)
irmsd_vals.extend([ir_fwd, ir_rev])
best_dockq = max(dockq_vals)
best_irmsd = min(irmsd_vals)
```

• Model B (alt partner naming): `1ahwBC_1jptH_0_1tfhB_0.rosetta_0001_0001.pdb` -> model chains `H/B`, native pairs `HT,LT`.

Plots

Generated plot files:

• `benchmark/prism_processed_results/benchmark_reports/plots/rigid_status_bar.png`

• `benchmark/prism_processed_results/benchmark_reports/plots/rigid_dockq_histogram.png`

• `benchmark/prism_processed_results/benchmark_reports/plots/rigid_irmsd_histogram.png`

• `benchmark/prism_processed_results/benchmark_reports/plots/rigid_dockq_vs_irmsd.png`

• `benchmark/prism_processed_results/benchmark_reports/plots/medium_status_bar.png`

• `benchmark/prism_processed_results/benchmark_reports/plots/medium_dockq_histogram.png`

• `benchmark/prism_processed_results/benchmark_reports/plots/medium_irmsd_histogram.png`

• `benchmark/prism_processed_results/benchmark_reports/plots/medium_dockq_vs_irmsd.png`

• `benchmark/prism_processed_results/benchmark_reports/plots/difficult_status_bar.png`

• `benchmark/prism_processed_results/benchmark_reports/plots/difficult_dockq_histogram.png`

• `benchmark/prism_processed_results/benchmark_reports/plots/difficult_irmsd_histogram.png`

• `benchmark/prism_processed_results/benchmark_reports/plots/difficult_dockq_vs_irmsd.png`

Plot confirmation notes:

• Per-set status bars use `Matched`, `DockQ Scored`, `iRMSD Scored`, `DockQ Errors`, `iRMSD Errors` from each set summary.

• Histograms and scatter use only rows with numeric metric values in `per_prediction.csv`.

Rigid Set Details

• Pair-summary rows: `110`

• DockQ scored rows: `378` (mean = `0.214`, max = `0.859`)

• iRMSD scored rows: `374` (mean = `11.358`, min = `0.000`)

• DockQ CAPRI classes: `Acceptable=43, High=11, Incorrect=250, Medium=74`

• DockQ failure reasons: `No interface found in native for selected mapping=1, Other DockQ error=20`

• iRMSD failure reasons: `Other iRMSD error=20, iRMSD script runtime failure=5`

Top 5 pairs by best DockQ:

• `2nnr|3bpf` (`2OUL_A:B`): best DockQ = `0.859`, n = `1`

• `1a19|1rgh` (`1AY7_A:B`): best DockQ = `0.854`, n = `1`

• `3pc6|3pc7` (`3PC8_A:C`): best DockQ = `0.843`, n = `6`

• `1ccp|1ycc` (`2PCC_A:B`): best DockQ = `0.840`, n = `3`

• `1c3d|2gom` (`3D5S_A:C`): best DockQ = `0.834`, n = `2`

Top 5 pairs by best (lowest) iRMSD:

• `1u5y|1xut` (`1XU1_ABD:T`): best iRMSD = `0.000`, n = `2`

• `3pc6|3pc7` (`3PC8_A:C`): best iRMSD = `0.600`, n = `6`

• `1a19|1rgh` (`1AY7_A:B`): best iRMSD = `0.612`, n = `1`

• `1c3d|2gom` (`3D5S_A:C`): best iRMSD = `0.642`, n = `2`

• `3f5v|3rvt` (`3RVW_CD:A`): best iRMSD = `0.642`, n = `1`

Medium Set Details

• Pair-summary rows: `34`

• DockQ scored rows: `120` (mean = `0.101`, max = `0.713`)

• iRMSD scored rows: `118` (mean = `13.507`, min = `0.000`)

• DockQ CAPRI classes: `Acceptable=16, Incorrect=96, Medium=8`

• DockQ failure reasons: `Other DockQ error=14`

• iRMSD failure reasons: `Other iRMSD error=14, iRMSD script runtime failure=2`

Top 5 pairs by best DockQ:

• `2g75|2ghv` (`2DD8_HL:S`): best DockQ = `0.713`, n = `2`

• `1iam|1mq9` (`1MQ8_A:B`): best DockQ = `0.587`, n = `1`

• `1l2z|1qgv` (`1SYX_A:B`): best DockQ = `0.584`, n = `1`

• `1hpt|2cga` (`1CGI_E:I`): best DockQ = `0.539`, n = `1`

• `1n0v|1xk9` (`1ZM4_A:B`): best DockQ = `0.531`, n = `2`

Top 5 pairs by best (lowest) iRMSD:

• `1h0c|2c0m` (`3R9A_AC:B`): best iRMSD = `0.000`, n = `2`

• `2g75|2ghv` (`2DD8_HL:S`): best iRMSD = `0.573`, n = `2`

• `1auq|1fvu` (`1IJK_A:BC`): best iRMSD = `1.712`, n = `2`

• `1mjn|3hi5` (`3HI6_XY:B`): best iRMSD = `1.867`, n = `1`

• `1jb1|2hpr` (`1KKL_ABC:H`): best iRMSD = `1.933`, n = `1`

Difficult Set Details

• Pair-summary rows: `17`

• DockQ scored rows: `67` (mean = `0.193`, max = `0.701`)

• iRMSD scored rows: `67` (mean = `13.004`, min = `1.088`)

• DockQ CAPRI classes: `Acceptable=22, Incorrect=41, Medium=4`

Top 5 pairs by best DockQ:

• `1jpe|1jzo` (`1JZD_AB:C`): best DockQ = `0.701`, n = `8`

• `1a6z|1cx8` (`1DE4_AB:CF`): best DockQ = `0.555`, n = `2`

• `1qfk|1tfh` (`1FAK_HL:T`): best DockQ = `0.476`, n = `3`

• `3h13|4jj7` (`3H11_BC:A`): best DockQ = `0.466`, n = `20`

• `1qup|2jcw` (`1JK9_B:A`): best DockQ = `0.458`, n = `2`

Top 5 pairs by best (lowest) iRMSD:

• `1jpe|1jzo` (`1JZD_AB:C`): best iRMSD = `1.088`, n = `8`

• `1fnl|3ave` (`1E4K_AB:C`): best iRMSD = `1.428`, n = `4`

• `1a6z|1cx8` (`1DE4_AB:CF`): best iRMSD = `2.513`, n = `2`

• `1qfk|1tfh` (`1FAK_HL:T`): best iRMSD = `2.569`, n = `3`

• `1buy|1ern` (`1EER_A:BC`): best iRMSD = `2.841`, n = `3`

Failure-Reason Notes

• `DockQ` failures are mostly due to: `No interface found in native for selected mapping`.

• This means the requested chain mapping exists syntactically, but DockQ cannot detect an interface in the native structure for that chain pair under its criteria.

• `iRMSD` failures are fewer and usually stem from chain/alignment/runtime issues for specific rows.

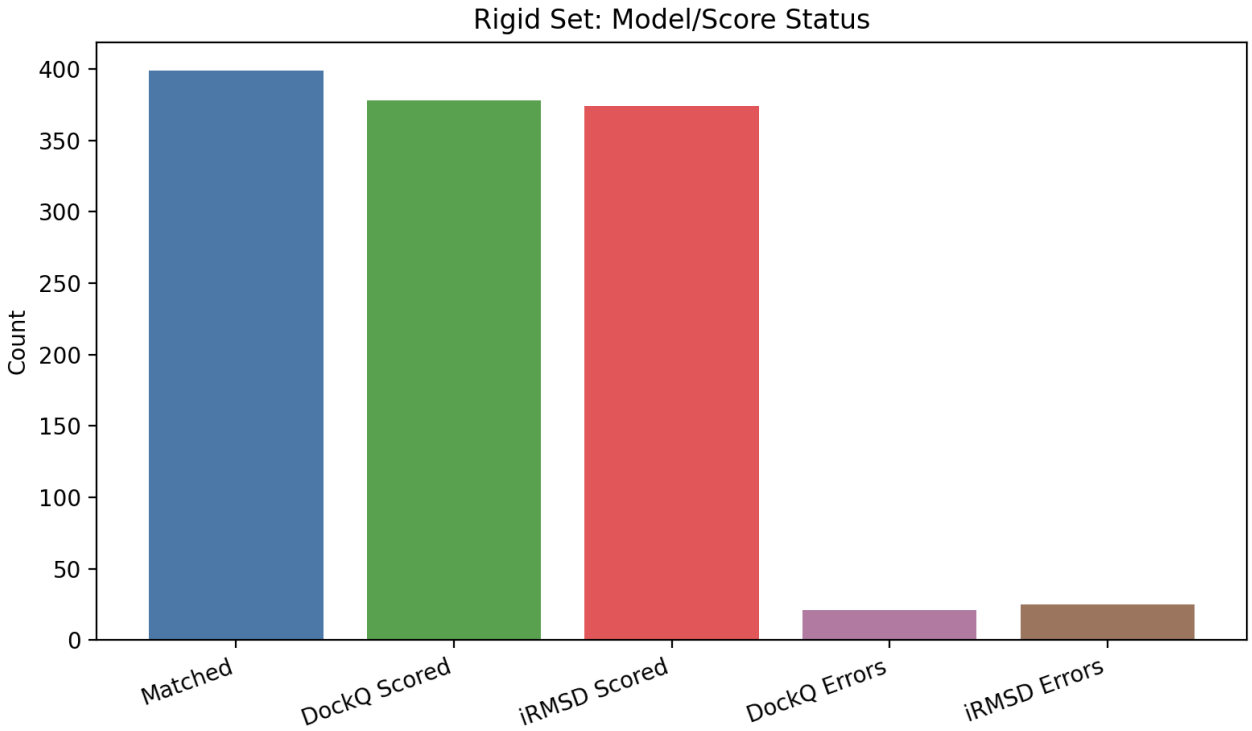
• Full error lists are exported to: `benchmark/prism_processed_results/benchmark_reports/error_files.csv`.

• Example error: set=`rigid`, metric=`dockq`, file=`2wp3OT_1tfhB_0_1jptL_0.rosetta_0001_0001.pdb`, complex=`1JPS_HL:T`, reason=`model_chain_missing_in_pdb:B`.

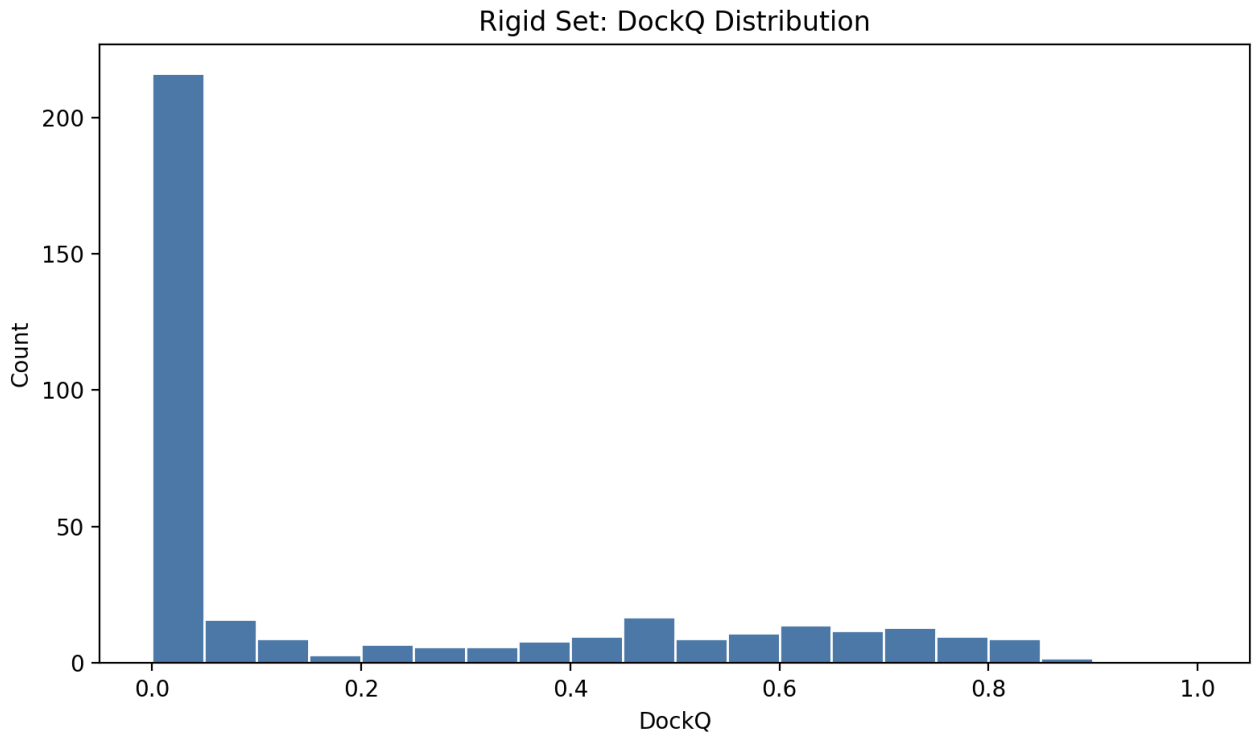
Figures

Rigid Set Figures

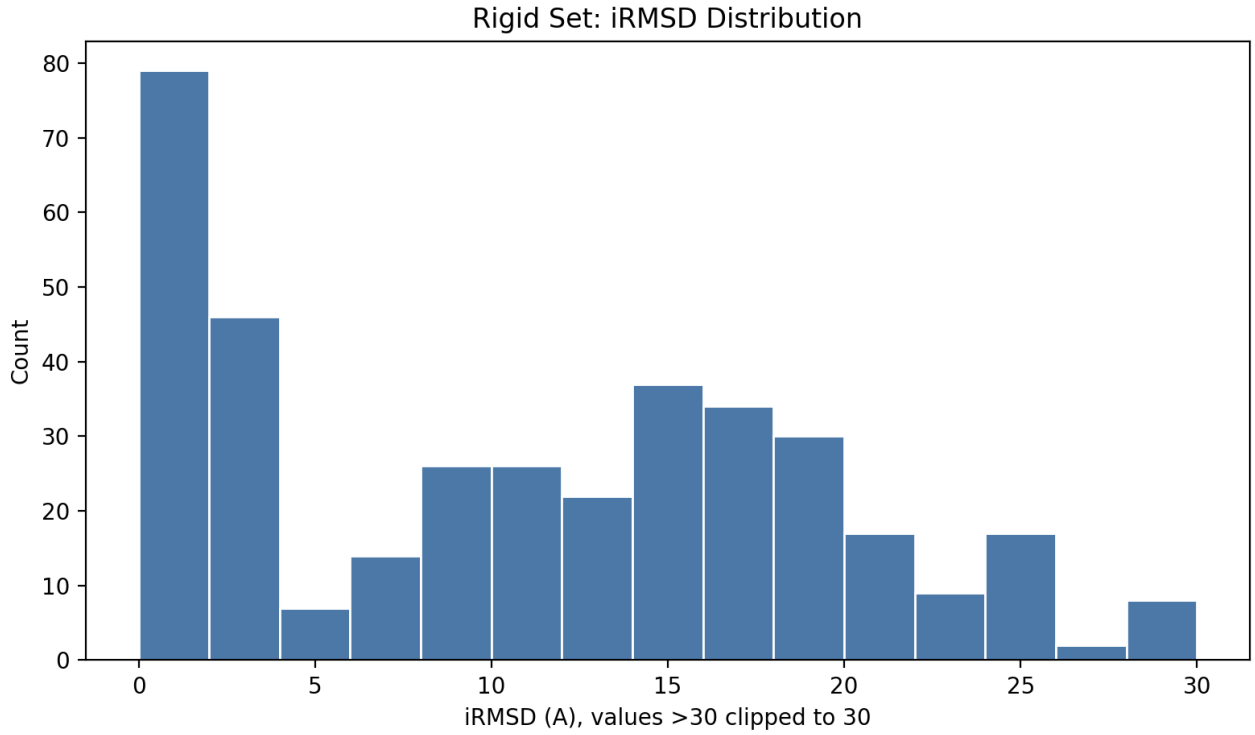
Rigid Status



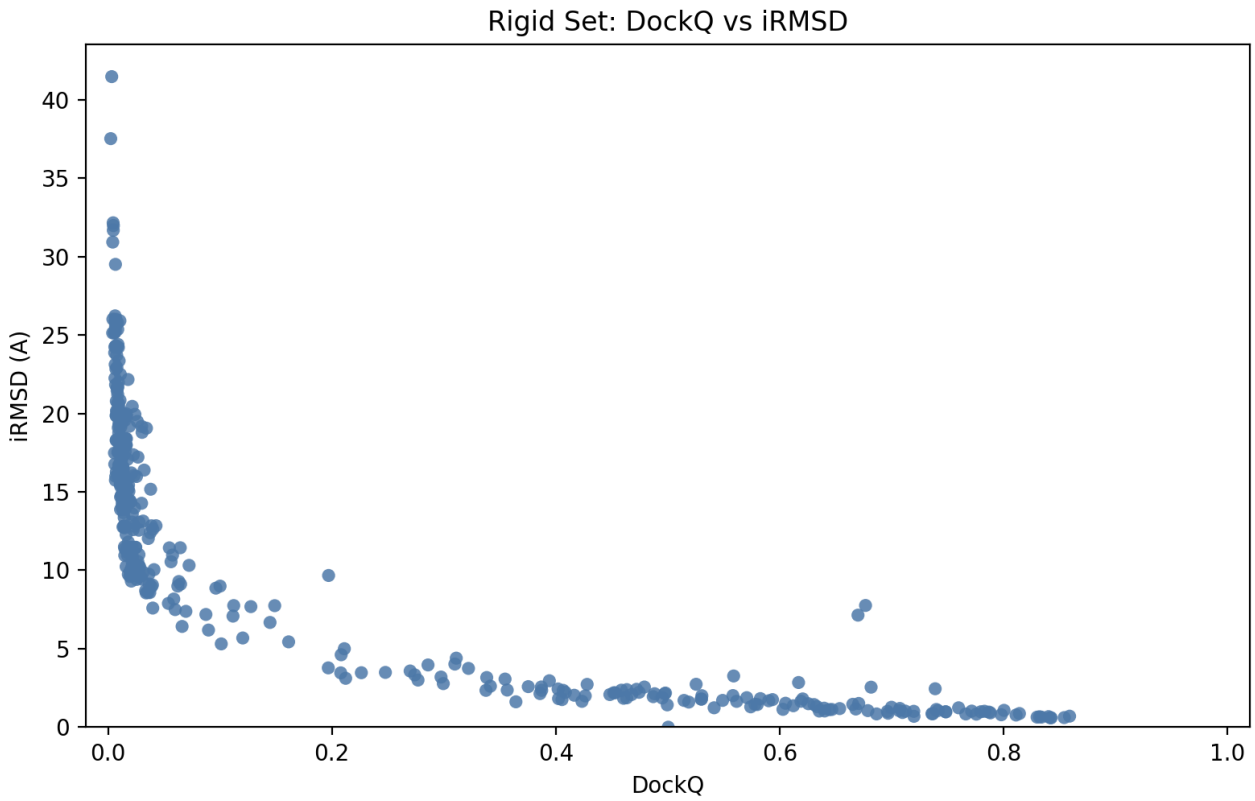
Rigid DockQ Distribution



Rigid iRMSD Distribution

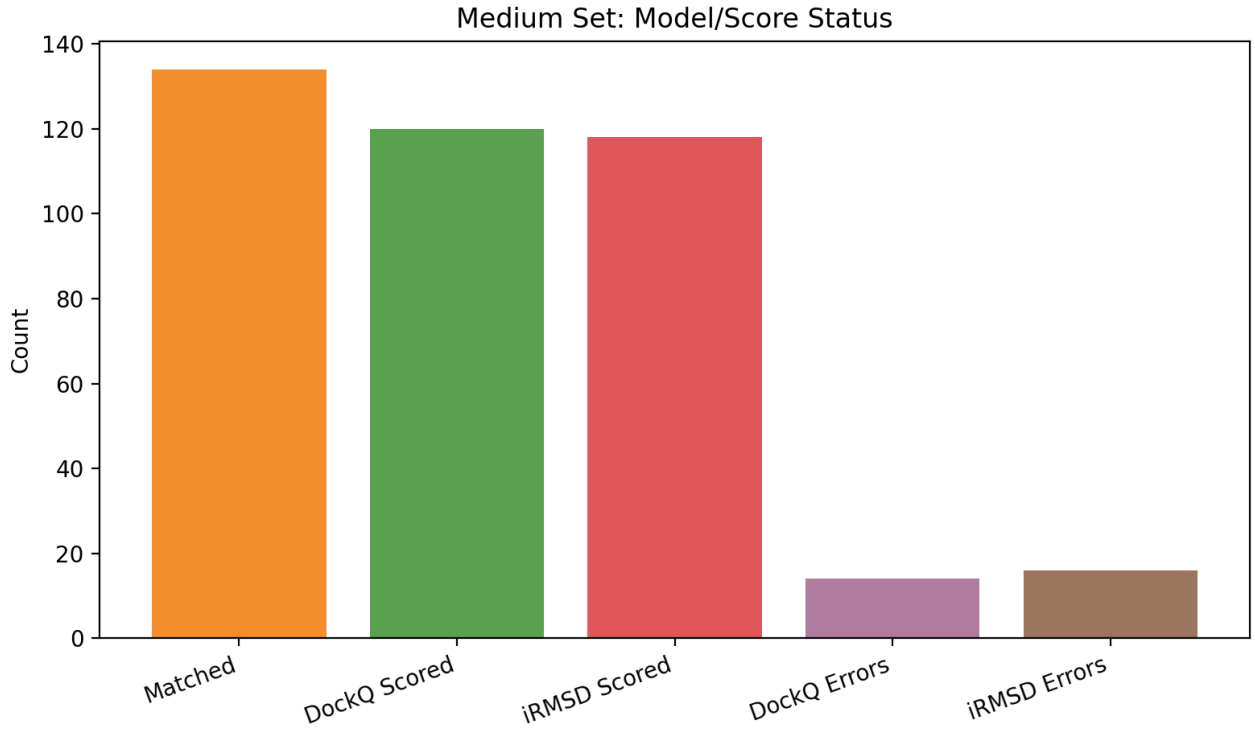


Rigid DockQ vs iRMSD

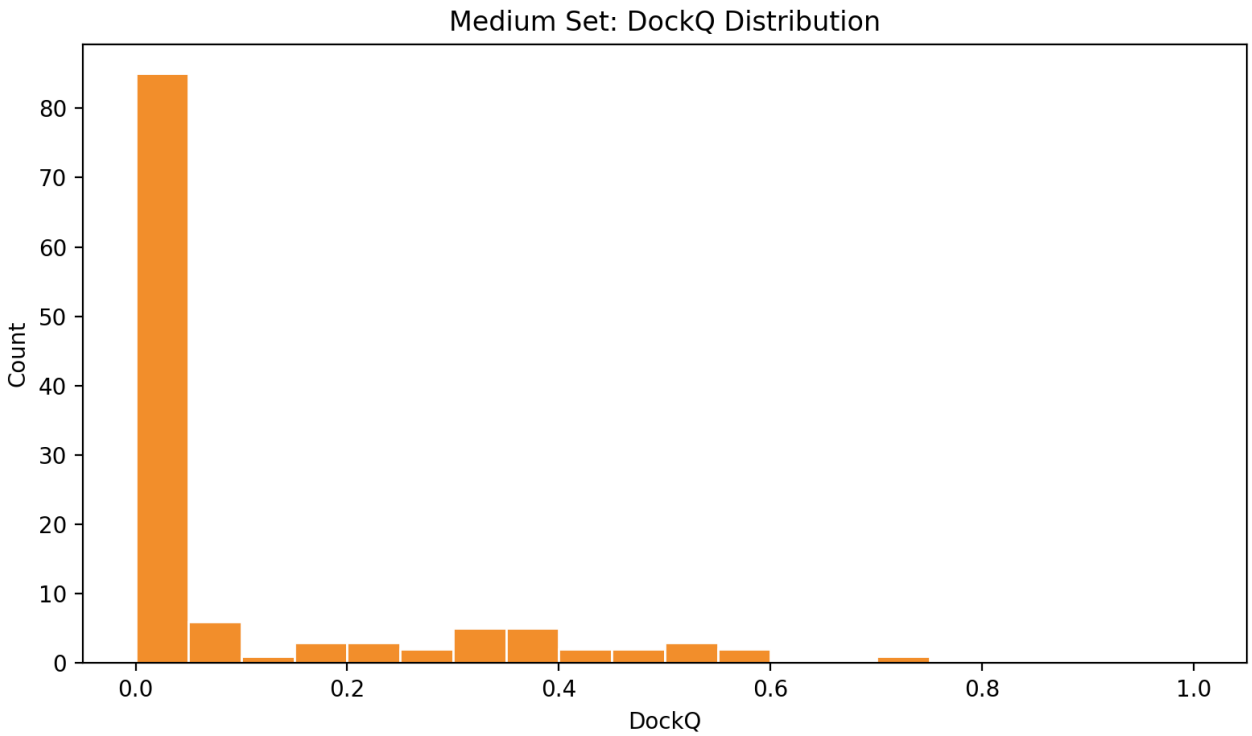


Medium Set Figures

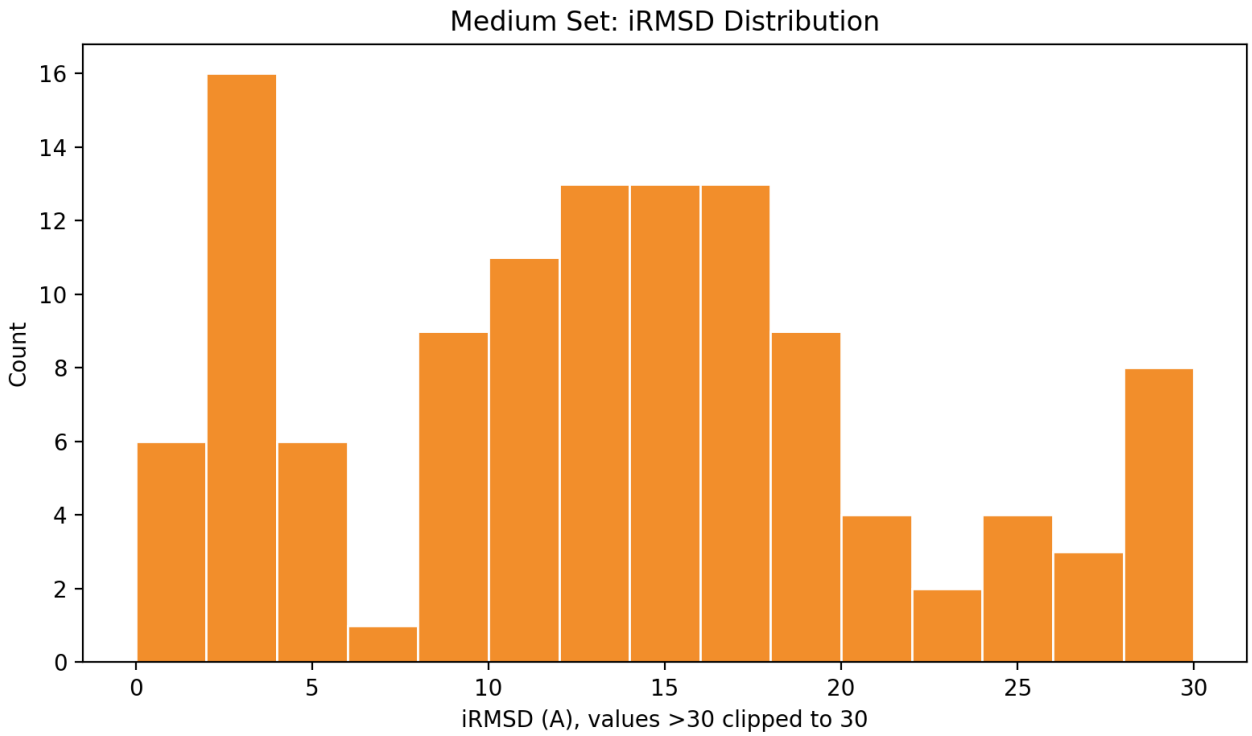
Medium Status



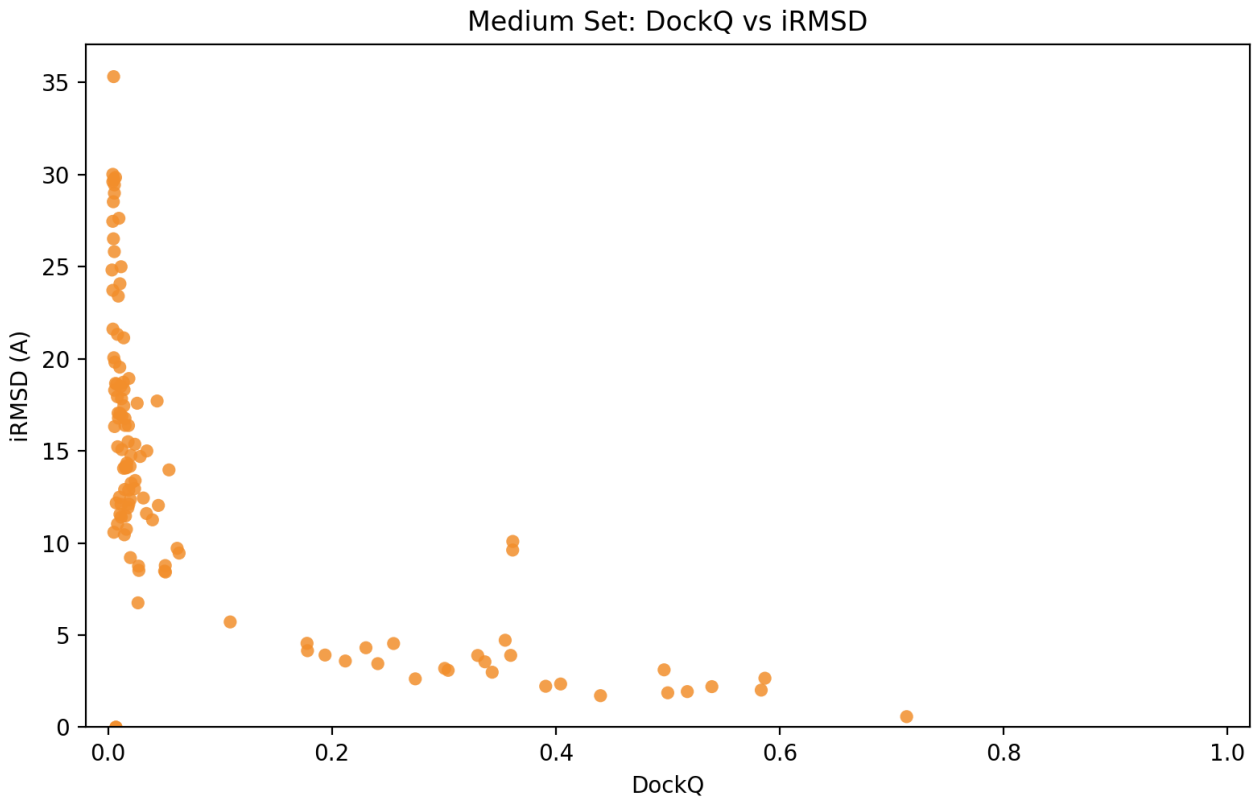
Medium DockQ Distribution



Medium iRMSD Distribution

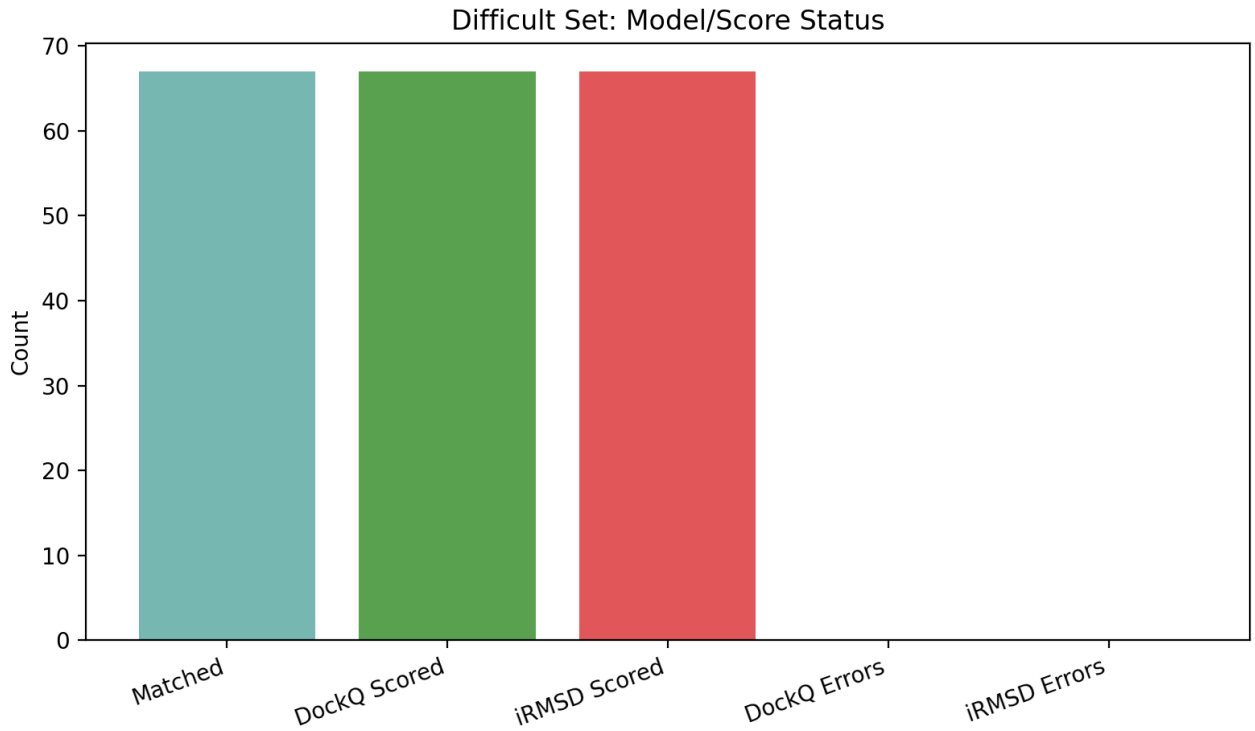


Medium DockQ vs iRMSD

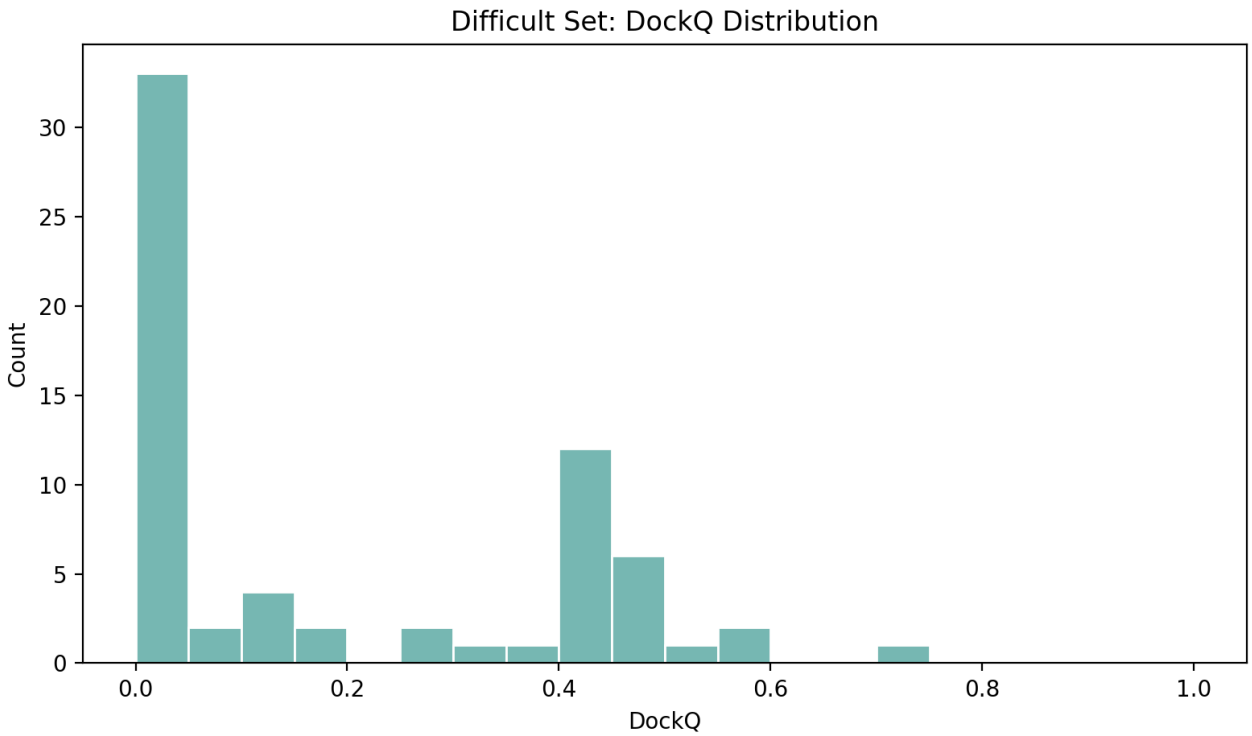


Difficult Set Figures

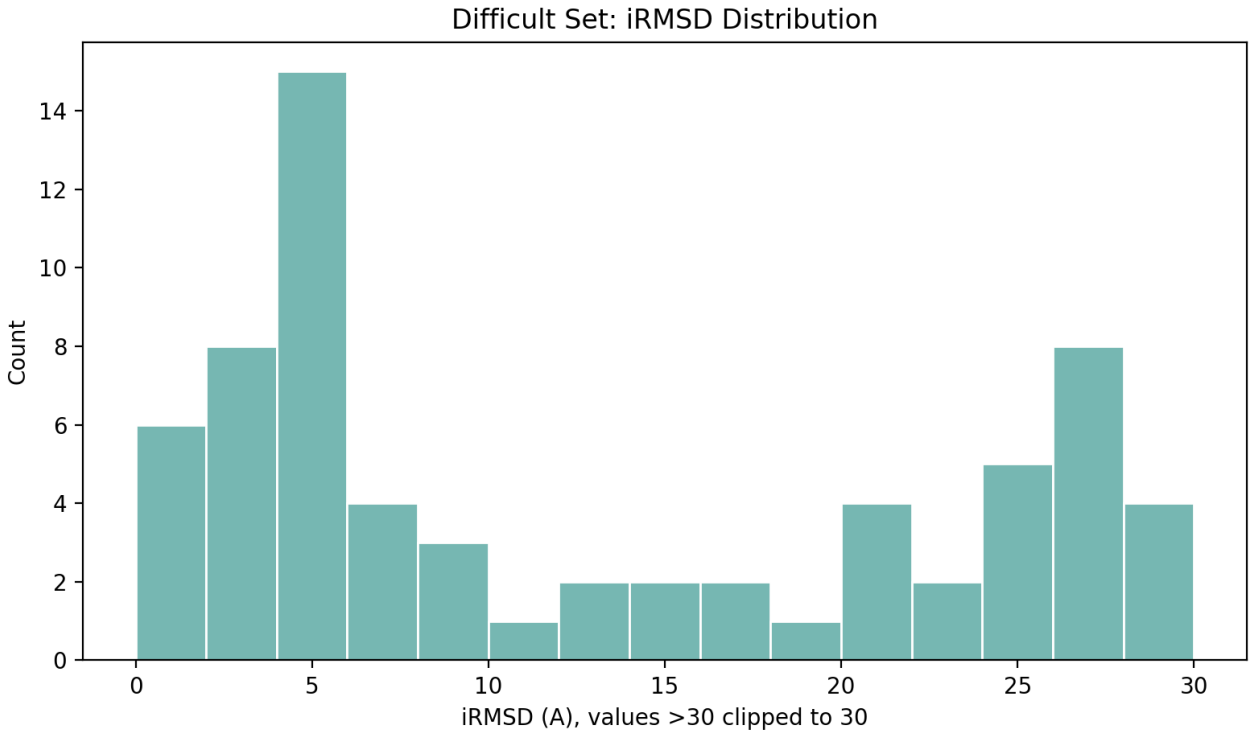
Difficult Status



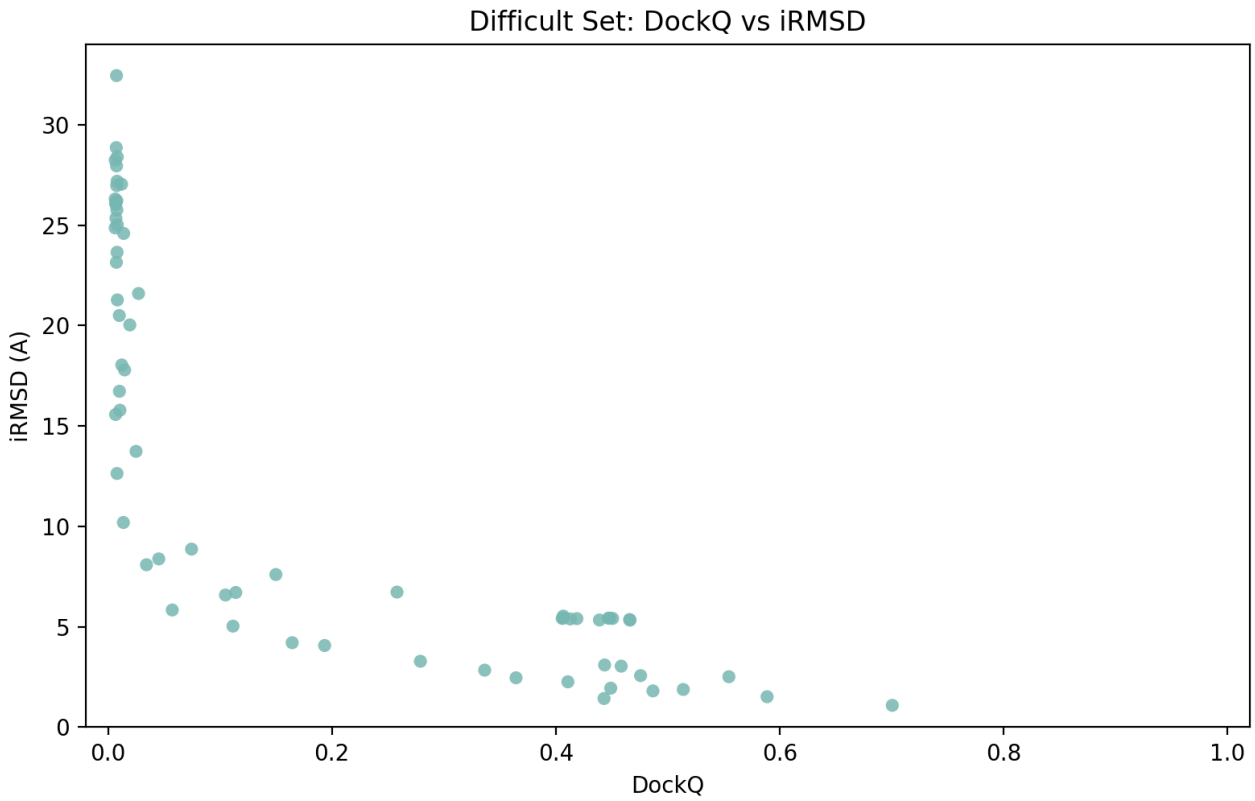
Difficult DockQ Distribution



Difficult iRMSD Distribution



Difficult DockQ vs iRMSD


Difficult Set: DockQ vs iRMSD
iRMSD (A)
DockQ
0
5
10
15
20
25
30
0.0
0.2
0.4
0.6
0.8
1.0